6 Related Work

To our knowledge, this is the first work to study inadvertent context leakage as an independent phenomenon. We position our findings with respect to prior work as follows.

Prompt injection. Prompt injection attacks manipulate LLM behavior by inserting malicious instructions into model inputs (Perez and Ribeiro, 2022). Greshake et al. (2023) extended this to *indirect* prompt injection via adversarial content in external sources, and Christodorescu et al. (2026) systematized overt exfiltration channels in agentic systems. Unlike these works, which focus on hijacking model behavior or exfiltrating data through explicit channels, our threat model considers an adversary who extracts private information through *covert* semantic channels.

Training-time and multi-agent threats. Data poisoning attacks manipulate training data to induce targeted misbehavior (Wan et al., 2023; Bagdasaryan and Shmatikov, 2022), backdoors can persist through safety training (Hubinger et al., 2024; Chaudhari et al., 2025), and steganographic encoders can be embedded via fine-tuning (Meier et al., 2025) or multi-agent collusion (Motwani et al., 2024; Cloud et al., 2025). All of these require training-time access or coordination between multiple models; we study what can be extracted from a single, unmodified model at inference time via its natural outputs.

Language model inversion. Model inversion attacks reconstruct private inputs from model outputs. Morris et al. (2024) showed model outputs can be inverted to recover input prompts; Zhang et al. (2022) demonstrated similar attacks on classifiers; and Song and Raghunathan (2020) and Li et al. (2023) showed that sentence embeddings reveal sensitive attributes. These attacks typically require access to model logits or embeddings; our adversary observes only the text output (i.e., the information exposed by frontier APIs).

Privacy in language models. Several frameworks address LLM privacy: contextual-integrity-based violation detection (Li et al., 2025; Shvartzshnaider and Duddu, 2025), privacy norm evaluation for agents (Shao et al., 2024), fine-tuning for contextual privacy (Xiao et al., 2024), and user disclosure risk studies (Zhang et al., 2024). These works focus on preventing *voluntary* disclosure; we study settings where the model discloses nothing explicitly, yet information leaks through the statistical properties of benign outputs.

Privacy in in-context learning. A parallel line studies privacy for in-context learning itself. Closest to us, Choi et al. (2025) audit privacy-preserving ICL methods: they insert identifiable canary tokens into prompt exemplars, craft queries to surface them, and compare the resulting worst-case leakage against a mechanism's theoretical privacy budget. Both papers ask what escapes a model's context, and the two approaches are complementary. In their threat model, a model that refuses the extraction request has protected the data, and that refusal is where our work begins.

Latent knowledge. Treutlein et al. (2024) showed that LLMs infer latent structure from disparate training documents, Betley et al. (2025) demonstrated behavioral self-awareness, and Cywiński et al. (2025) studied eliciting knowledge that models possess but refuse to verbalize. These works suggest that LLMs maintain latent representations that could surface in output distributions even when the model refuses to state them explicitly, motivating our study of in-context secret leakage.

7 Conclusion

We have shown that in-context secrets are not safe: even when a language model correctly refuses to reveal a secret, its benign outputs carry nonzero information about the secret's value. The phenomenon extends to natural language, where semantic predicates about user memories are inferrable from routine task outputs. This passive leakage enables active exploitation: an RL-trained adversary extracts full Social Security Numbers from production-style agents powered by two frontier models (Claude Opus 4.6 and Gemini 3.1 Pro). Our findings indicate that defending in-context secrets requires more than alignment training and output filtering; information-theoretic guarantees on the generation process may be necessary.

References

Anthropic. System card: Claude opus 4.6. Technical report, Anthropic, February 2026a. https://www-cdn.anthropic.com/6a5fa276ac68b9aeb0c8b6af5fa36326e0e166dd.pdf.

Anthropic. System card: Claude sonnet 4.6. Technical report, Anthropic, February 2026b. https://www-cdn.anthropic.com/bbd8ef16d70b7a1665f14f306ee88b53f686aa75.pdf.

Eugene Bagdasaryan and Vitaly Shmatikov. Spinning language models: Risks of propaganda-as-a-service and countermeasures. In *2022 IEEE Symposium on Security and Privacy (SP)*, 2022. doi: 10.1109/SP46214.2022.9833572.

Jan Betley, Xuchan Bao, Martín Soto, Anna Sztyber-Betley, James Chua, and Owain Evans. Tell me about yourself: LLMs are aware of their learned behaviors. In *The Thirteenth International Conference on Learning Representations*, 2025. https://openreview.net/forum?id=IjQ2Jtemzy.

Harsh Chaudhari, Jamie Hayes, Matthew Jagielski, Ilia Shumailov, Milad Nasr, and Alina Oprea. Cascading adversarial bias from injection to distillation in language models. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, 2025. doi: 10.1145/3719027.3765122.

Jacob Choi, Shuying Cao, Xingjian Dong, and Sai Praneeth Karimireddy. ContextLeak: Auditing leakage in private in-context learning methods. In *The Impact of Memorization on Trustworthy Foundation Models: ICML 2025 Workshop*, 2025. https://openreview.net/forum?id=IwSJz7y7Pg.

Mihai Christodorescu, Earlence Fernandes, Ashish Hooda, Somesh Jha, Johann Rehberger, Kamalika Chaudhuri, Xiaohan Fu, Khawaja Shams, Guy Amir, Jihye Choi, et al. Systems security foundations for agentic computing, 2026. https://arxiv.org/abs/2512.01295.

Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv preprint arXiv:2507.14805*, 2025.

Bartosz Cywiński, Emil Ryd, Rowan Wang, Senthooran Rajamanoharan, Neel Nanda, Arthur Conmy, and Samuel Marks. Eliciting secret knowledge from language models. *arXiv preprint arXiv:2510.01070*, 2025.

Fyxer AI. Fyxer AI, 2026. https://www.fyxer.com/. Accessed: 2026-05-07.

Google DeepMind. Gemini 3.1 flash lite model card. Technical report, Google DeepMind, March 2026a. https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Flash-Lite-Model-Card.pdf.

Google DeepMind. Gemini 3.1 pro model card. Technical report, Google DeepMind, February 2026b. https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Pro-Model-Card.pdf.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models, 2024. https://arxiv.org/abs/2407.21783.

Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've signed up for: Compromising real-world llm-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security*, 2023.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 2025.

Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M Ziegler, Tim Maxwell, Newton Cheng, et al. Sleeper agents: Training deceptive llms that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024.

Haoran Li, Mingshi Xu, and Yangqiu Song. Sentence embedding leaks more information than you expect: Generative embedding inversion attack to recover the whole sentence. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. https://aclanthology.org/2023.findings-acl.881/.

Haoran Li, Wei Fan, Yulin Chen, Cheng Jiayang, Tianshu Chu, Xuebing Zhou, Peizhao Hu, and Yangqiu Song. Privacy checklist: Privacy violation detection grounding on contextual integrity theory. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2025. https://aclanthology.org/2025.naacl-long.86/.

Dominik Meier, Jan Philip Wahle, Paul Röttger, Terry Ruas, and Bela Gipp. Trojanstego: Your language model can secretly be a steganographic privacy leaking agent. *arXiv preprint arXiv:2505.20118*, 2025.

Meta. Muse and spark safety and preparedness report. Technical report, Meta, April 2026. https://ai.meta.com/static-resource/muse-spark-safety-and-preparedness-report/.

Niloofar Mireshghallah, Hyunwoo Kim, Xuhui Zhou, Yulia Tsvetkov, Maarten Sap, Reza Shokri, and Yejin Choi. Can llms keep a secret? testing privacy implications of language models via contextual integrity theory. *arXiv preprint arXiv:2310.17884*, 2023.

Niloofar Mireshghallah, Neal Mangaokar, Narine Kokhlikyan, Arman Zharmagambetov, Manzil Zaheer, Saeed Mahloujifar, and Kamalika Chaudhuri. CIMemories: A compositional benchmark for contextual integrity in LLMs. In *The Fourteenth International Conference on Learning Representations*, 2026. https://openreview.net/forum?id=YnNIp38v1M.

John Xavier Morris, Wenting Zhao, Justin T Chiu, Vitaly Shmatikov, and Alexander M Rush. Language model inversion. In *The Twelfth International Conference on Learning Representations*, 2024. https://openreview.net/forum?id=t9dWHpGkPj.

Sumeet Motwani, Mikhail Baranchuk, Martin Strohmeier, Vijay Bolina, Philip Torr, Lewis Hammond, and Christian Schroeder de Witt. Secret collusion among ai agents: Multi-agent deception via steganography. In *Advances in Neural Information Processing Systems*, 2024.

Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, et al. Olmo 3, 2026. https://arxiv.org/abs/2512.13961.

OpenAI. Introducing GPT-5.4 mini and nano. https://openai.com/index/introducing-gpt-5-4-mini-and-nano/, March 2026a. Accessed: 2026-05-06.

OpenAI. GPT-5.4 Thinking System Card. https://deploymentsafety.openai.com/gpt-5-4-thinking, March 2026b. Accessed: 2026-05-06.

Fábio Perez and Ian Ribeiro. Ignore previous prompt: Attack techniques for language models. *arXiv preprint arXiv:2211.09527*, 2022.

Qwen Team. Qwen2.5: A party of foundation models, September 2024. https://qwenlm.github.io/blog/qwen2.5/.

Yijia Shao, Tianshi Li, Weiyan Shi, Yanchen Liu, and Diyi Yang. Privacylens: Evaluating privacy norm awareness of language models in action. In *Advances in Neural Information Processing Systems*, 2024. https://proceedings.neurips.cc/paper_files/paper/2024/file/a2a7e58309d5190082390ff10ff3b2b8-Paper-Datasets_and_Benchmarks_Track.pdf.

Yan Shvartzshnaider and Vasisht Duddu. Position: Contextual integrity is inadequately applied to language models. In *42nd International Conference on Machine Learning Position Paper Track*, 2025. https://openreview.net/forum?id=YmTxiR1HUX.

Congzheng Song and Ananth Raghunathan. Information leakage in embedding models. In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, 2020.

Peter Steinberger. OpenClaw: Personal AI assistant, 2025. https://github.com/openclaw/openclaw. Software, MIT License.

Johannes Treutlein, Dami Choi, Jan Betley, Samuel Marks, Cem Anil, Roger B Grosse, and Owain Evans. Connecting the dots: Llms can infer and verbalize latent structure from disparate training data. In *Advances in Neural Information Processing Systems*, 2024.

Alexander Wan, Eric Wallace, Sheng Shen, and Dan Klein. Poisoning language models during instruction tuning. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. https://proceedings.mlr.press/v202/wan23b.html.

Yuxin Wen, Arman Zharmagambetov, Ivan Evtimov, Narine Kokhlikyan, Tom Goldstein, Kamalika Chaudhuri, and Chuan Guo. Rl is a hammer and llms are nails: A simple reinforcement learning recipe for strong prompt injection, 2025. https://arxiv.org/abs/2510.04885.

xAI. Grok 4 model card. Technical report, xAI, August 2025a. https://data.x.ai/2025-08-20-grok-4-model-card.pdf.

xAI. Grok 4.1 model card. Technical report, xAI, November 2025b. https://data.x.ai/2025-11-17-grok-4-1-model-card.pdf.